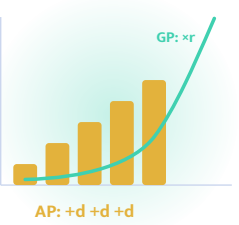
MT Classes, Motihari

Predicting What Comes Next

Class 9 Maths, Chapter 8: Exploring Sequences and Progressions (NCERT Ganita Manjari, Part I)



Number patterns that let you see the future. Learn explicit and recursive rules, APs and GPs, Gauss's pairing trick and the magic of fractals, with number machines, a growing-pattern builder, a fractal zoomer, a bouncing ball and every exercise solved.

8.1 What is a sequence?

Patterns are everywhere: in nature, art, music and money. Patterns help us **predict what comes next**. In maths, a **sequence** is an **ordered list of numbers**, and each number in it is called a **term**.

Natural numbers

1, 2, 3, 4, 5, 6, ...

+1 each time

Odd numbers

1, 3, 5, 7, 9, 11, ...

+2 each time

Triangular numbers

1, 3, 6, 10, 15, 21, ...

+2, +3, +4, +5, +6

Square numbers

1, 4, 9, 16, 25, 36, ...

+3, +5, +7, +9, +11

The three dots **...** mean the sequence goes on forever. Such a sequence is **infinite**. A list that stops, such as 6, 12, 24, 48, 96 (five terms), is a **finite** sequence. Roll numbers in your class, or the dates of the Mondays in a month, are finite sequences from daily life.

🤔 **Think and Reflect: Describe the pattern in each sequence above and predict the next few numbers.**

Natural numbers: add 1 → 7, 8, 9. Odd numbers: add 2 → 13, 15, 17. Triangular numbers: the gap grows by 1 each time (+7, +8, +9) → 28, 36, 45. Square numbers: the gap is the next odd number (+13, +15, +17) → 49, 64, 81.

Triangular numbers are sums

1 = 1, 3 = 1 + 2, 6 = 1 + 2 + 3, 10 = 1 + 2 + 3 + 4, 15 = 1 + 2 + 3 + 4 + 5.

So each **triangular number** is the **sum of the natural numbers up to that term**. Stack the dots in rows of 1, 2, 3, ... and you get a triangle.

The next two pictures would have 21 dots (6 rows) and 28 dots (7 rows).

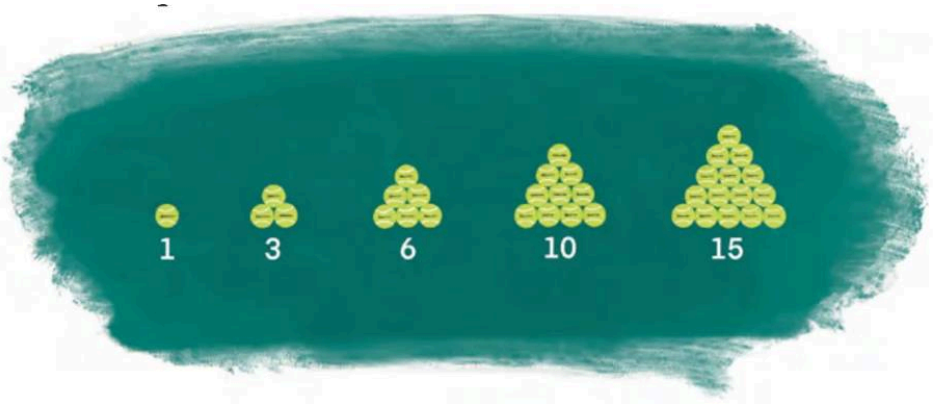


Fig. 8.1 The first five triangular numbers.

Square numbers are sums of odd numbers

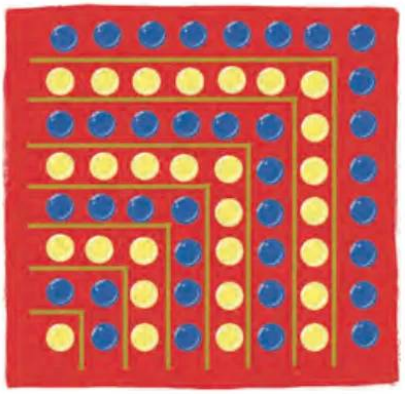


Fig. 8.2 Square numbers and odd numbers.

1 = 1, 4 = 1 + 3, 9 = 1 + 3 + 5, 16 = 1 + 3 + 5 + 7, ...

In Fig. 8.2 each new **L-shaped layer** (alternately blue and yellow) wraps around the square before it. The layers have 1, 3, 5, 7, ... dots, so

1 + 3 + 5 + ... (n odd numbers) = n^2

– **Ex** **Predict the next four terms of 1, 4, 7, 10, 13, ... and find the first 10 terms of the sequence of running totals.**

The gap is always 3, so the next four terms are **16, 19, 22, 25**.

Running totals: 1, 1+4 = 5, 5+7 = 12, 12+10 = 22, 22+13 = 35, 35+16 = 51, 51+19 = 70, 70+22 = 92, 92+25 = 117, 117+28 = 145.

First 10 terms: **1, 5, 12, 22, 35, 51, 70, 92, 117, 145**.

A short way to name terms: t_n

🔑 Notation

t_1 = first term, t_2 = second term, ..., **t_n = the term in position n**. The small number (subscript) is the *position*, not the value.

For odd numbers: $t_1 = 1$, $t_2 = 3$, $t_3 = 5$, $t_4 = 7$. So $t_4 = 7$ means "the 4th term is 7". For different sequences we use different letters: t_n, s_n, u_n, ...

position n	1	2	3	4	5

term t_n	1	3	5	7	9

Terms can be fractions, like 1, $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, ... (decreasing), or negative, like −7, −3, 1, 5, 9, ... (here $s_1 = -7$ and the gap is 4). The **position n** is always a counting number, but the **term** can be any real number.

– **Ex** **Write t_5, t_6, t_7 and t_8 for the triangular numbers.**

$t_5 = 15$, $t_6 = 15 + 6 =$ **21**, $t_7 = 21 + 7 =$ **28**, $t_8 = 28 + 8 =$ **36**.

🤔 **Think and Reflect: List five different kinds of sequences.**

For example: (1) multiples of 5: 5, 10, 15, ... (increasing, equal gaps); (2) a countdown 10, 9, 8, ... (decreasing); (3) powers of 2: 1, 2, 4, 8, ... (each term doubles); (4) prime numbers 2, 3, 5, 7, 11, ... (no simple rule); (5) alternating 1, −1, 1, −1, ... (repeats).

8.2 Explicit rule: a formula for t_n

An **explicit formula** uses the **position number n** to calculate the term directly. Put in n, get out t_n, like a machine.



Example 1: $u_n = 2n - 1$

$u_1 = 2\times1 - 1 = 1$, $u_2 = 2\times2 - 1 = 3$, $u_3 = 2\times3 - 1 = 5$, ... These are the **odd numbers**.

Why is it useful? You can jump straight to the 20th, 53rd or 300th term without knowing the terms before it.

– **Ex** **Using $u_n = 2n - 1$, find the 53rd, 108th and 1170th odd numbers.**

$u_{53} = 106 - 1 =$ **105**; $u_{108} = 216 - 1 =$ **215**; $u_{1170} = 2340 - 1 =$ **2339**.

Is a number a term? Solve for n

To check if a number belongs to the sequence, set t_n equal to it and solve. **If n comes out as a natural number (1, 2, 3, ...), it is a term. Otherwise it is not**, because there is no "94.6th position".

✅

Is 137 an odd number term?

$2n - 1 = 137 \Longrightarrow 2n = 138 \Longrightarrow n = 69$. Yes, the **69th** term.

❓

$s_n = 5n - 2$: 308 and 471?

$5n - 2 = 308 \Longrightarrow n = 62$ ✅ (62nd term).

$5n - 2 = 471 \Longrightarrow n = 94.6$ ❌ not a term.

– **Ex 2** **Write the first 6 terms of $s_n = 5n - 2$. What are the 100th and 1000th terms?**

3, 8, 13, 18, 23, 28. $s_{100} = 500 - 2 =$ **498**; $s_{1000} = 5000 - 2 =$ **4998**.

🤔 **Think and Reflect: What is the rule for the nth square number? Is there a rule for primes?**

Square numbers: $\mathbf{t_n = n^2}$. The primes 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 have **no simple formula**; nobody has found a rule that predicts the next prime easily. That is what makes them so interesting!

– **Ex** **For $t_n = 3n - 7$: (i) find $t_1, t_2, t_3, t_{12}, t_{18}, t_{50}$ (ii) which term is 332? (iii) is 557 a term?**

(i) $t_1 = -4, t_2 = -1, t_3 = 2, t_{12} = 29, t_{18} = 47, t_{50} = 143$.

(ii) $3n - 7 = 332 \Longrightarrow 3n = 339 \Longrightarrow n = 113$. So 332 is the **113th term**.

(iii) $3n - 7 = 557 \Longrightarrow 3n = 564 \Longrightarrow n = 188$, a natural number. **Yes, 557 is the 188th term.**

8.3

Recursive rule: each term from the one before

In 1, 4, 7, 10, 13, … each term is 3 more than the previous one. We can write this as

$$\mathbf{t_1 = 1,\ \ t_n = t_{n-1} + 3 \ \text{ for } n \geq 2}$$

t_{n-1} means "the term just before t_n". The explicit rule for the same sequence is $t_n = 3n - 2$.

A **recursive rule** describes a term using **earlier terms**. It always needs a **starting value**, and to find the 50th term you must walk through all the terms before it.

🎯

Explicit

Jump straight to any term: $t_n = 3n - 2$.

Like knowing a house number.

🪜

Recursive

Climb step by step: $t_n = t_{n-1} + 3$.

Like following directions "next door".

Example 3: $u_1 = 1, u_n = 2u_{n-1} + 3$

$u_2 = 2\times1 + 3 = 5$, $u_3 = 2\times5 + 3 = 13$, $u_4 = 2\times13 + 3 = 29$. First four terms: **1, 5, 13, 29**.

Is 133 a term? Keep going: $u_5 = 61$, $u_6 = 125$, $u_7 = 253$. The terms jump from 125 to 253, so **133 is not a term**.

Example 4: $s_1 = 3, s_n = s_{n-1}(s_{n-1} - 1)$

$s_2 = 3 \times 2 = 6$, $s_3 = 6 \times 5 = 30$, $s_4 = 30 \times 29 = 870$. First four terms: **3, 6, 30, 870**. Notice how fast it grows!

The Virahaṅka–Fibonacci sequence

A recursive rule can use the previous **two** (or more) terms. The most famous one is

$$V_1 = 1, V_2 = 2, \quad V_n = V_{n-1} + V_{n-2} \text{ for } n \geq 3$$

1, 2, 3, 5, 8, 13, 21, 34, 55, 89

Each term is the sum of the previous two. The next two terms after 34 are **55** and **89**.

🇮🇳 **An Indian discovery**

This sequence was first written down by **Virahānka** in the 7th century CE in his book *Vṛttajātisamuchaya*, while counting rhythms in Prakrit poetry. It was studied further by **Gopāla** (c. 1135 CE) and **Hemachandra** (c. 1150 CE), and only later by the Italian mathematician **Fibonacci** (c. 1200 CE).

Ex 8.1

Exercise Set 8.1: solutions

Try each question first, then tap to check.

1. **Find the first five terms: (i) $t_n = 3n - 4$, (ii) $t_n = 2 - 5n$, (iii) $t_n = n^2 - 2n + 3$.**

n	1	2	3	4	5
$3n - 4$	−1	2	5	8	11
$2 - 5n$	−3	−8	−13	−18	−23
$n^2 - 2n + 3$	2	3	6	11	18

For (iii), e.g. $t_4 = 16 - 8 + 3 = 11$ and $t_5 = 25 - 10 + 3 = 18$.

2. **Find the 10th and 15th terms of $t_n = 5n - 3$.**

$t_{10} = 50 - 3 =$ **47**; $t_{15} = 75 - 3 =$ **72**.

3. **Are 97 and 172 terms of $t_n = 5n - 3$?**

$5n - 3 = 97 \implies 5n = 100 \implies n = 20$ ✅. **97 is the 20th term.**

$5n - 3 = 172 \implies 5n = 175 \implies n = 35$ ✅. **172 is the 35th term.**

4. **Which term of $t_n = 5n - 3$ is 607?**

$5n - 3 = 607 \implies 5n = 610 \implies n = 122$. **607 is the 122nd term.**

5. **$t_1 = -5$, $t_{n+1} = t_n + 3$. Find the first five terms. Is 52 a term?**

Terms: **−5, −2, 1, 4, 7**. Each term adds 3, so the explicit rule is $t_n = -5 + 3(n - 1) = 3n - 8$.

$3n - 8 = 52 \implies 3n = 60 \implies n = 20$. **Yes, 52 is the 20th term.**

6. **$T_1 = 1$, $T_2 = 2$, $T_3 = 4$, $T_n = T_{n-1} + T_{n-2} + T_{n-3}$. Find T_4 to T_8.**

$T_4 = 4 + 2 + 1 =$ **7**, $T_5 = 7 + 4 + 2 =$ **13**, $T_6 = 13 + 7 + 4 =$ **24**, $T_7 = 24 + 13 + 7 =$ **44**, $T_8 = 44 + 24 + 13 =$ **81**.

8.4 Arithmetic progressions (AP)

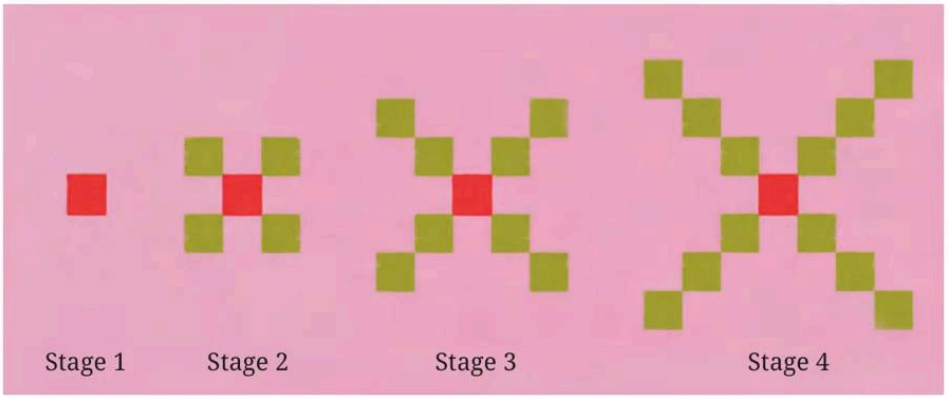


Fig. 8.3 Growing pattern of squares: 1, 5, 9, 13, ...

At each stage **4 squares are added**, one at each corner. So the counts are

$$\mathbf{1,\ 1+1\times4,\ 1+2\times4,\ 1+3\times4,\ \ldots \Longrightarrow\ t_n = 1+(n-1)\times4 = 4n-3}$$

🔑 Arithmetic progression

A sequence in which the **difference between consecutive terms is constant**. This constant is the **common difference d**.

$$\mathbf{a,\ a+d,\ a+2d,\ a+3d,\ \ldots,\ a+(n-1)d}$$

$$\mathbf{t_n = a+(n-1)d}$$

a = first term, d = common difference = $t_2 - t_1$. d can be positive, negative or zero.

Recursive: $\mathbf{t_1 = a,\ t_n = t_{n-1} + d}$

1, 4, 7, 10, ...
a = 1, d = 3
$t_n = 1 + (n-1)\times3$

11, 7, 3, −1, −5, ...
a = 11, d = −4
decreasing AP

1, 5, 9, 13, ...
a = 1, d = 4
$t_n = 4n - 3$

🤔 **Think and Reflect: Which sequences in this chapter are APs?**

APs: natural numbers (d = 1), odd numbers (d = 2), 1, 4, 7, 10, ... (d = 3), −7, −3, 1, 5, ... (d = 4), 5n − 2, 3n − 7, the squares pattern 1, 5, 9, 13 (d = 4).

Not APs: triangular numbers and square numbers (gaps keep growing), 1, ½, ⅓, ... (gaps shrink), primes, Virahānka numbers, and 6, 12, 24, 48, 96 (it doubles).

8.4.1 Visualising an AP: a straight line!

Stage	1	2	3	4	5	n
Squares	1	5	9	13	17	4n − 3

Plot the pairs (stage, squares): (1, 1), (2, 5), (3, 9), (4, 13), (5, 17). **They lie on a straight line.** This happens for every AP, because each step right goes up by the same amount d.

📈 AP ⟹ points on a **straight line**. The steepness of the line is the common difference d.

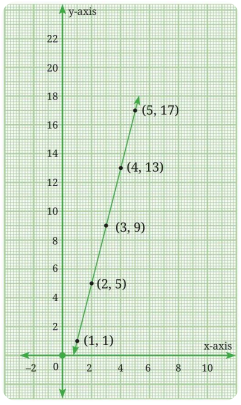


Fig. 8.4 An AP plotted: a linear pattern.

– **Ex** **Show that (i) 2, 5, 8, 11, ... and (ii) −5, −1, 3, 7, ... are APs and find their nth terms.**

(i) Differences are all 3, so it is an AP with a = 2, d = 3: $t_n = 2 + (n - 1)3 = \mathbf{3n - 1}$.

(ii) Differences are all 4, so a = −5, d = 4: $t_n = -5 + (n - 1)4 = \mathbf{4n - 9}$.

Plotting (n, t_n) gives points on a straight line in both cases.

Ex **Find the nth term of (i) $\frac{1}{2}, \frac{5}{2}, \frac{9}{2}, \frac{13}{2}, \ldots$ (ii) 1.5, 3.5, 5.5, 7.5, ...**

(i) $a = \frac{1}{2}, d = 2$: $t_n = \frac{1}{2} + 2(n-1) = \mathbf{2n - \frac{3}{2}}$ (that is, $\frac{4n-3}{2}$).

(ii) $a = 1.5, d = 2$: $t_n = 1.5 + 2(n-1) = \mathbf{2n - 0.5}$.

Ex **Write recursive rules for the APs above.**

2, 5, 8, ...: $t_1 = 2, t_n = t_{n-1} + 3$. −5, −1, 3, ...: $t_1 = -5, t_n = t_{n-1} + 4$.
$\frac{1}{2}, \frac{5}{2}, \ldots$: $t_1 = \frac{1}{2}, t_n = t_{n-1} + 2$. 1.5, 3.5, ...: $t_1 = 1.5, t_n = t_{n-1} + 2$.

🚕 Example 5: Taxi fare

Booking fee ₹200 plus ₹40 per km. Fare after 1, 2, 3 km: **240, 280, 320, ...**, an AP with $a = 240, d = 40$.

$t_n = 240 + (n-1)\times 40 = \mathbf{200 + 40n}$. For 10 km: 200 + 400 = **₹600**.

8.5 Sum of the first n natural numbers

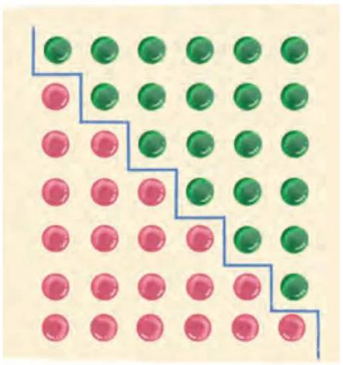
Can you add 1 + 2 + ... + 10 without adding one by one? Write the sum **forwards and backwards**:

S =	1	2	3	4	5	6	7	8	9	10
S =	10	9	8	7	6	5	4	3	2	1
2S =	11	11	11	11	11	11	11	11	11	11

Every column adds to 11, and there are 10 columns. So 2S = 10 × 11 = 110, and **S = 55**.

The same trick works for any n: S = 1 + 2 + ... + n and S = n + (n − 1) + ... + 1. Each column adds to (n + 1), and there are n columns:

$$\mathbf{2S = n(n+1) \implies S_n = 1 + 2 + \ldots + n = \frac{n(n+1)}{2}}$$



$2 \times (1 + 2 + 3 + 4 + 5 + 6) = 7 \times 6$

Fig. 8.5 $2 \times (1 + 2 + 3 + 4 + 5 + 6) = 7 \times 6$

The picture proof. The pink staircase has $1 + 2 + \ldots + 6$ circles. The green staircase is the same staircase turned upside down. Together they make a **rectangle of $7 \times 6 = 42$** circles, so one staircase has $42 \div 2 = 21$.

📜 Āryabhaṭa knew it

The first known written form of this result is in **Āryabhaṭa's *Āryabhaṭīya*** (Chapter 2, Verse 19). It says: add the first and last terms, halve it (the average), and multiply by the number of terms.

🤔 **Think and Reflect: Find S_{20}, S_{50}, S_{100} and S_{1000}.**

$S_{20} = \frac{20 \times 21}{2} = \mathbf{210}$; $S_{50} = \frac{50 \times 51}{2} = \mathbf{1275}$; $S_{100} = \frac{100 \times 101}{2} = \mathbf{5050}$; $S_{1000} = \frac{1000 \times 1001}{2} = \mathbf{500500}$.

Sum of a block of numbers: 25 + 26 + … + 58

Take everything up to 58 and remove everything up to 24:

$S_{58} - S_{24} = \frac{58 \times 59}{2} - \frac{24 \times 25}{2} = 1711 - 300 = \mathbf{1411}$.

🤔 **Think and Reflect: Triangular numbers are $t_n = \frac{n(n+1)}{2}$. Find the 10th, 17th and 80th.**

$t_{10} = \frac{10 \times 11}{2} = \mathbf{55}$; $t_{17} = \frac{17 \times 18}{2} = \mathbf{153}$; $t_{80} = \frac{80 \times 81}{2} = \mathbf{3240}$.

Ex 8.2

Exercise Set 8.2: solutions

1. Find the 10th and 26th terms of the AP 3, 8, 13, 18, ...

$a = 3, d = 5$. $t_{10} = 3 + 9\times5 =$ **48**; $t_{26} = 3 + 25\times5 =$ **128**.

2. Which term of 21, 18, 15, ... is −81? Is 0 a term?

$a = 21, d = -3$, so $t_n = 21 - 3(n - 1) = 24 - 3n$.

$24 - 3n = -81 \Longrightarrow 3n = 105 \Longrightarrow n = 35$. **−81 is the 35th term.**

$24 - 3n = 0 \Longrightarrow n = 8$, a natural number. **Yes, 0 is the 8th term** (21, 18, 15, 12, 9, 6, 3, 0).

3. Find the nth term of 11, 8, 5, 2, ... and its recursive rule.

$a = 11, d = -3$: $t_n = 11 - 3(n - 1) =$ **$14 - 3n$**.

Recursive: **$t_1 = 11, t_n = t_{n-1} - 3$** for $n \geq 2$.

4. An AP has 50 terms; the 3rd term is 12 and the last term is 106. Find the 29th term.

$a + 2d = 12$ and $a + 49d = 106$. Subtracting: $47d = 94 \Longrightarrow$ **$d = 2$**, so $a = 12 - 4 =$ **8**.

$t_{29} = 8 + 28\times2 =$ **64**.

5. How many 2-digit numbers are divisible by 3? What is their sum?

They are 12, 15, 18, ..., 99: an AP with $a = 12, d = 3$. $12 + (n - 1)3 = 99 \Longrightarrow n - 1 = 29 \Longrightarrow$ **$n = 30$**.

Sum $= 3 \times (4 + 5 + \ldots + 33) = 3 \times (S_{33} - S_3) = 3 \times (561 - 6) =$ **1665**.

Check with Āryabhaṭa's rule: $\frac{30}{2} \times (12 + 99) = 15 \times 111 = 1665$.

6. Harish's salary starts at ₹5,00,000 and rises by ₹20,000 each year. When does it reach ₹7,00,000?

Salaries form an AP: 5,00,000, 5,20,000, 5,40,000, ... with $d = 20{,}000$.

$5{,}00{,}000 + (n - 1)\times20{,}000 = 7{,}00{,}000 \Longrightarrow n - 1 = 10 \Longrightarrow n = 11$.

So in his **11th year**, that is **after 10 years** (10 increments), his salary is ₹7,00,000.

– **7. Marbles in rows of 1, 2, 3, …, 25. How many marbles in all?**

$S_{25} = \frac{25 \times 26}{2}$ = **325 marbles**.

8.6 Geometric progressions (GP)

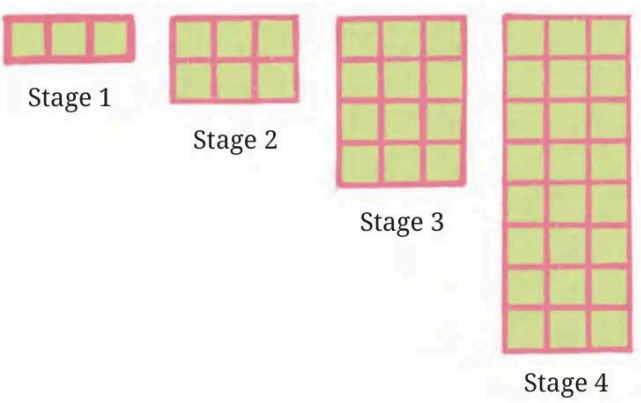


Fig. 8.6 A growing pattern: 3, 6, 12, 24, …

Here the number of squares **doubles** each stage: 3, 3 + 3 = 6, 6 + 6 = 12, 12 + 12 = 24.

3, 3×2, 3×2², 3×2³, … so $\mathbf{t_n = 3 \times 2^{n-1}}$.

Recursive: $t_1 = 3$, $t_n = 2t_{n-1}$.

The ratio of each term to the one before is always the same: $\frac{6}{3} = \frac{12}{6} = \frac{24}{12} = \frac{48}{24} = \frac{96}{48} = 2$. This constant multiplier is the **common ratio r**.

🔑 Geometric progression

A sequence where each term is the previous term **multiplied by a fixed number r**.

a, ar, ar², ar³, …, ar^{n-1}

$t_n = ar^{n-1}$

a = first term, r = common ratio = $\frac{t_2}{t_1}$

Recursive: $t_1 = a$, $t_n = r \times t_{n-1}$

AP: add the same number

Check the **difference** $t_2 - t_1, t_3 - t_2, \ldots$

Grows steadily (in a straight line).

GP: multiply by the same number

Check the **ratio** $\frac{t_2}{t_1}, \frac{t_3}{t_2}, \ldots$

Grows (or shrinks) faster and faster.

Ex 6: 1, 2, 4, 8, 16, ...
GP, r = 2

Ex 7: 1, 3, 9, 27, 81, ...
GP, r = 3

Ex 8: 1, −1, 1, −1, ...
GP, r = −1

Example 9: Is 5, $\frac{15}{4}, \frac{45}{16}, \frac{135}{64}, \ldots$ a GP?

$\frac{t_2}{t_1} = \frac{15}{4} \div 5 = \frac{3}{4}$; $\frac{t_3}{t_2} = \frac{45}{16} \times \frac{4}{15} = \frac{3}{4}$; $\frac{t_4}{t_3} = \frac{135}{64} \times \frac{16}{45} = \frac{3}{4}$.

The ratio is constant, so it is a GP with a = 5, r = $\frac{3}{4}$: $\mathbf{t_n = 5 \times (\frac{3}{4})^{n-1}}$.

– **Ex** **Are these GPs? Find t_n: (i) 2, 10, 50, 250, ... (ii) 4, $\frac{8}{3}, \frac{16}{9}, \frac{32}{27}, \ldots$ (iii) 3, $\frac{-3}{2}, \frac{3}{4}, \frac{-3}{8}, \ldots$**

(i) Ratio 5 each time. GP, $\mathbf{t_n = 2 \times 5^{n-1}}$.

(ii) Ratio $\frac{2}{3}$ each time. GP, $\mathbf{t_n = 4 \times (\frac{2}{3})^{n-1}}$.

(iii) Ratio $\frac{-1}{2}$ each time. GP, $\mathbf{t_n = 3 \times (\frac{-1}{2})^{n-1}}$. A negative ratio makes the signs alternate.

– **Ex** **Give a recursive rule for $t_n = 3 \times 10^{n-1}$ (3, 30, 300, 3000, ...).**

$\mathbf{t_1 = 3, t_n = 10 \times t_{n-1}}$ for $n \geq 2$.

8.6.1 Fun with fractals

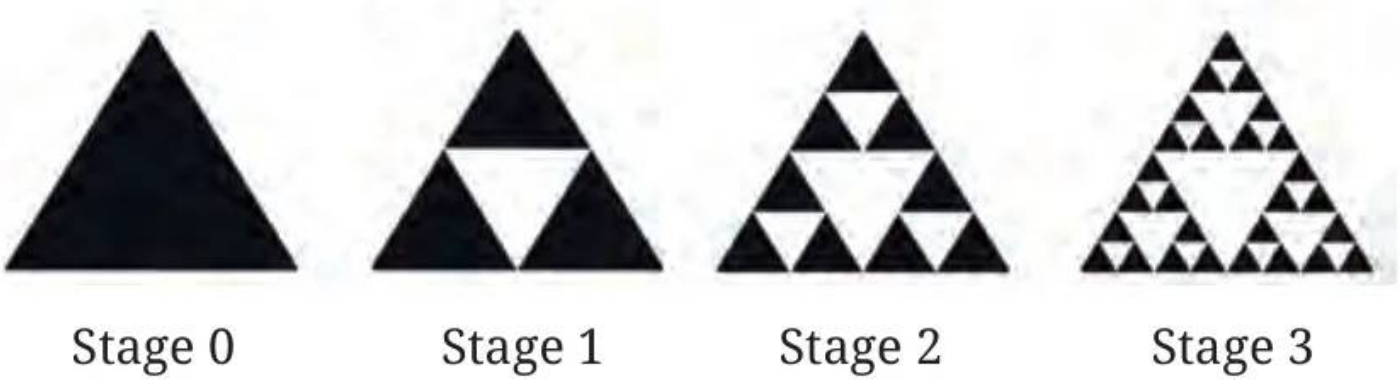


Fig. 8.7 The Sierpiński triangle, stages 0 to 3.

1. Stage 0: a black equilateral triangle.
2. Join the midpoints of the sides to get 4 smaller triangles and **remove the middle one** (Stage 1).
3. Repeat on every black triangle that is left, again and again, forever!

🔺

Number of black triangles

1, 3, 9, 27, 81, 243, ... each one becomes 3.

GP with r = 3: $\mathbf{t_n = 3^n}$ (stage n).

⬛

Black area (Stage 0 = 1)

1, $\frac{3}{4}$, $\frac{9}{16}$, $\frac{27}{64}$, ... each stage keeps $\frac{3}{4}$.

GP with r = $\frac{3}{4}$: $\mathbf{s_n = (\frac{3}{4})^n}$.

Stage n	0	1	2	3	4	5	n
Black triangles	1	3	9	27	81	243	3^n
Black area	1	$\frac{3}{4}$	$\frac{9}{16}$	$\frac{27}{64}$	$\frac{81}{256}$	$\frac{243}{1024}$	$(\frac{3}{4})^n$

🤯 **The surprise:** the number of triangles shoots up (243 at stage 5, over 59,000 at stage 10), but the black **area keeps shrinking towards 0**. Both are GPs: one with r > 1, one with r between 0 and 1.

🤔 **Think and Reflect: answers to (a) to (d) for the Sierpiński triangle**

(a) 1, 3, 9, 27. (b) Stage 4: 81, Stage 5: 243. (c) 3^n. (d) Areas: $\frac{3}{4}$, $\frac{9}{16}$, $\frac{27}{64}$; Stage 4: $\frac{81}{256}$, Stage 5: $\frac{243}{1024}$; rule $(\frac{3}{4})^n$. As n grows, the area gets closer and closer to 0. Recursive rules: $t_n = 3 \times t_{n-1}$ and $s_n = \frac{3}{4} \times s_{n-1}$, starting from 1 at stage 0.

Fractals are shapes that **repeat themselves at different scales**: zoom into a small part and it looks like the whole. You see them in tree branches, cauliflower and broccoli, snowflakes and coastlines. Simple rules, beautiful designs.

👤 Wacław Sierpiński (1882–1969)

A Polish mathematician who gave us the Sierpiński gasket (triangle), one of the earliest and most famous fractals.



Fig. 8.8 Fractals in nature.

8.6.2 Visualising a GP: a curve, not a line

Stage	1	2	3	4	5	n
Squares	3	6	12	24	48	$3\times2^{n-1}$

Plot (1, 3), (2, 6), (3, 12), (4, 24), (5, 48). The points **do not lie on a straight line**. They bend upwards more and more steeply, because each jump is bigger than the last.

📈 AP ⟹ straight line. 📈 GP ⟹ curve that bends up ($r > 1$) or flattens down towards 0 ($0 < r < 1$).

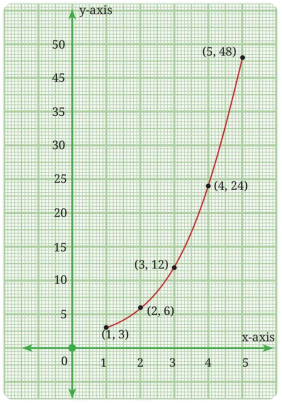


Fig. 8.9 Points of a GP do not lie on a straight line.

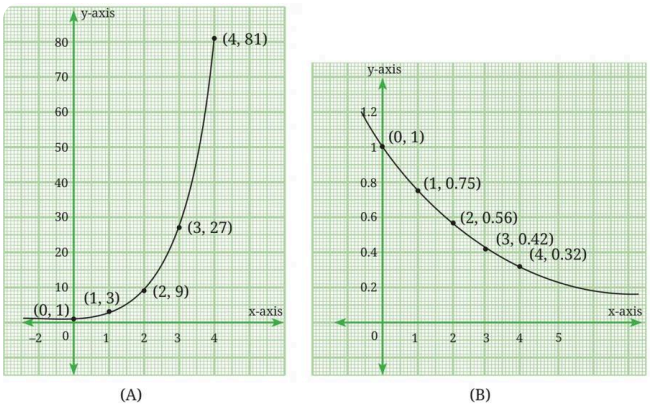


Fig. 8.10 Sierpiński triangle: (A) the number of triangles rises very quickly, (B) the black area falls closer and closer to 0.

Example 10: The bouncing ball

A ball is dropped from **24 feet**. Each bounce reaches $\frac{3}{4}$ of the previous height.

Bounce	Height (ft)
1st	24 × 0.75 = 18
2nd	18 × 0.75 = 13.5
3rd	13.5 × 0.75 = 10.125
4th	10.125 × 0.75 ≈ 7.594
5th	≈ 5.695
6th	≈ 4.271
7th	≈ 3.203

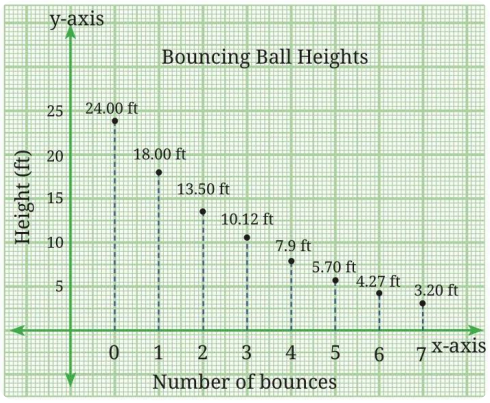


Fig. 8.11 Bouncing ball heights.

(a) The heights 18, 13.5, 10.125, 7.594, 5.695, ... form a GP with a = 18 and r = $\frac{3}{4}$.

(b) $\frac{1}{6}$ of 24 ft is **4 ft**. After the 6th bounce the ball still reaches 4.27 ft, but after the **7th bounce** it reaches only 3.20 ft. So **7 bounces** are needed.

Ex 8.3 Exercise Set 8.3: solutions

1. A GP has r = 2 and 8th term 192. Find the 12th term.

From the 8th to the 12th term we multiply by r four times: $t_{12} = 192 \times 2^4 = 192 \times 16 = \mathbf{3072}$.

2. Find the 10th and nth terms of 5, 25, 125, ...

$a = 5, r = 5$: $t_n = 5 \times 5^{n-1} = \mathbf{5^n}$. $t_{10} = 5^{10} = \mathbf{97{,}65{,}625}$.

3*. $t_1 = 2, t_{n+1} = 3t_n - 2$. Which term is 730?

Terms: 2, 3×2−2 = 4, 3×4−2 = 10, 28, 82, 244, 3×244−2 = **730**. That is the **7th term**.

Pattern spotted: each term is one more than a power of 3 ($2 = 3^0+1, 4 = 3^1+1, 10 = 3^2+1$, ...), so $t_n = 3^{n-1} + 1$ and $730 = 3^6 + 1$ gives n = 7.

4. Which term of 2, 6, 18, ... is 4374? Give explicit and recursive formulas.

$a = 2, r = 3$: $t_n = 2 \times 3^{n-1}$. $2 \times 3^{n-1} = 4374 \implies 3^{n-1} = 2187 = 3^7 \implies n = 8$. **4374 is the 8th term.**

Explicit: $\mathbf{t_n = 2 \times 3^{n-1}}$. Recursive: $\mathbf{t_1 = 2, t_n = 3t_{n-1}}$.

5. A ball dropped from 80 m bounces to 60% of its fall each time. (i) Height after the 5th bounce? (ii) Total vertical distance when it hits the ground the 6th time?

Bounce heights: 48, 28.8, 17.28, 10.368, 6.2208 m (GP, r = 0.6).

(i) $80 \times 0.6^5 = \mathbf{6.2208\ m \approx 6.22\ m}$.

(ii) First fall 80 m. Then each of the 5 bounces goes up and comes down again before the 6th hit: 2 × (48 + 28.8 + 17.28 + 10.368 + 6.2208) = 2 × 110.6688 = 221.3376 m.

Total = 80 + 221.3376 = **301.3376 m ≈ 301.34 m**.

– **6. Which term of 2, 2√2, 4, … is 128?**

$r = \sqrt{2}$. $t_n = 2(\sqrt{2})^{n-1} = 128 \Rightarrow (\sqrt{2})^{n-1} = 64 = 2^6 = (\sqrt{2})^{12} \Rightarrow n - 1 = 12 \Rightarrow \mathbf{n = 13}$.

– **7. Sierpiński square carpet (Fig. 8.12): red squares and red area at each stage.**

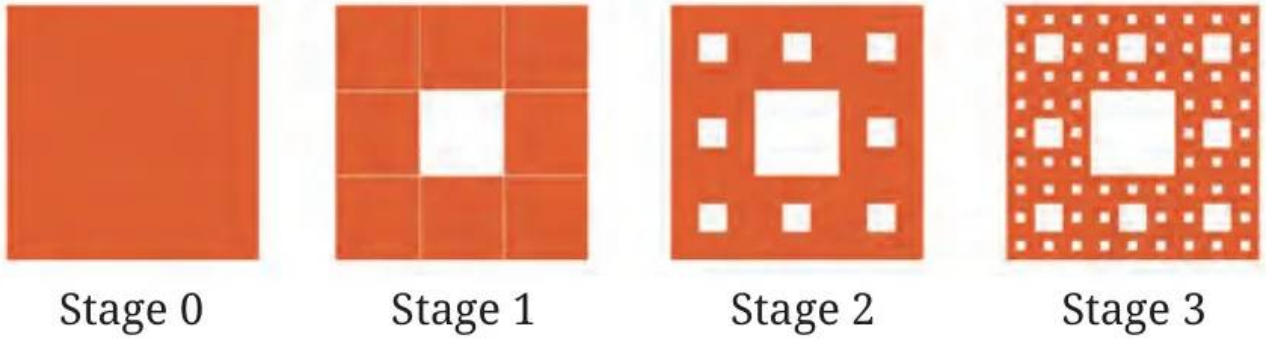


Fig. 8.12 Stages 0 to 3 of the Sierpiński square carpet.

Each square is cut into 9 and the middle one is removed, so every red square becomes **8** squares, and the area is multiplied by $\frac{8}{9}$.

(i) Stages 0 to 3: **1, 8, 64, 512** red squares.

(ii) Stage 4: $8^4 =$ **4096**; Stage 5: $8^5 =$ **32768**.

(iii) Explicit: $\mathbf{t_n = 8^n}$ (stage n). Recursive: $t_0 = 1$, $\mathbf{t_n = 8 \times t_{n-1}}$.

(iv) Areas: Stage 1: $\frac{8}{9}$, Stage 2: $\frac{64}{81}$, Stage 3: $\frac{512}{729}$, Stage 4: $\frac{4096}{6561}$, Stage 5: $\frac{32768}{59049}$. Explicit: $\mathbf{s_n = (\frac{8}{9})^n}$; recursive: $s_0 = 1$, $\mathbf{s_n = \frac{8}{9} \times s_{n-1}}$. As n grows, the red area gets closer and closer to **0** (slowly, because $\frac{8}{9}$ is close to 1).

End

End-of-chapter exercises

– **1. Find the 31st term of an AP whose 11th term is 38 and 16th term is 73.**

$a + 10d = 38$ and $a + 15d = 73$. Subtracting: $5d = 35 \Rightarrow d = 7$, so $a = 38 - 70 = -32$.

$t_{31} = -32 + 30 \times 7 =$ **178**.

2. Find the AP whose 3rd term is 16 and whose 7th term exceeds the 5th term by 12.

$t_7 - t_5 = 2d = 12 \implies d = 6$. $a + 2d = 16 \implies a = 4$.

The AP is **4, 10, 16, 22, 28, ...**

3*. How many three-digit numbers are divisible by 7?

Smallest: 105 (7×15). Largest: 994 (7×142). AP with $a = 105$, $d = 7$.

$105 + (n - 1)7 = 994 \implies (n - 1)7 = 889 \implies n - 1 = 127 \implies \mathbf{n = 128}$.

4*. How many multiples of 4 lie between 10 and 250?

From 12 to 248, with $d = 4$: $12 + (n - 1)4 = 248 \implies n - 1 = 59 \implies \mathbf{n = 60}$.

5*. Find a GP where the sum of the first two terms is −4 and the 5th term is 4 times the 3rd term.

$ar^4 = 4ar^2 \implies r^2 = 4 \implies r = 2$ or $r = -2$.

$a + ar = -4$. If $r = 2$: $3a = -4$, $a = \frac{-4}{3} \rightarrow \mathbf{\frac{-4}{3}, \frac{-8}{3}, \frac{-16}{3}, \ldots}$

If $r = -2$: $-a = -4$, $a = 4 \rightarrow$ **4, −8, 16, −32, ...** Both GPs work.

6*. Find all ways of writing 100 as a sum of consecutive natural numbers.

If k numbers start at m: sum $= \frac{k(2m + k - 1)}{2} = 100$, so $k(2m + k - 1) = 200$. Try the factors of 200 with $k \geq 2$:

- $k = 5$: $2m + 4 = 40 \implies m = 18 \rightarrow$ **18 + 19 + 20 + 21 + 22 = 100**
- $k = 8$: $2m + 7 = 25 \implies m = 9 \rightarrow$ **9 + 10 + 11 + 12 + 13 + 14 + 15 + 16 = 100**
- $k = 2, 4, 10, 20, 25, 40, \ldots$ give m that is not a natural number.

So there are **two ways** (three if you count "100" on its own).

7*. 30 bacteria double every hour. How many after 2 hours, 4 hours, n hours?

GP: 30, 60, 120, 240, 480, ... After 2 hours: **120**; after 4 hours: **480**; after n hours: $\mathbf{30 \times 2^n}$.

8*. In an AP, $t_4 + t_8 = 24$ and $t_6 + t_{10} = 44$. Find the first three terms.

$2a + 10d = 24$ and $2a + 14d = 44$. Subtracting: $4d = 20 \Rightarrow d = 5$, so $2a = -26$, $a = -13$.

First three terms: **−13, −8, −3**.

9*. Smallest n for which 1 + 2 + ... + n is greater than 1000.

We need $n(n + 1) > 2000$. $44 \times 45 = 1980$ ($S_{44} = 990$, too small) and $45 \times 46 = 2070$ ($S_{45} = 1035$). So **n = 45**.

10*. Which term of 2, 8, 32, ... is 131072? Give explicit and recursive formulas.

$a = 2$, $r = 4$: $2 \times 4^{n-1} = 131072 \Rightarrow 4^{n-1} = 65536 = 4^8 \Rightarrow$ **n = 9**.

Explicit: $\mathbf{t_n = 2 \times 4^{n-1}}$. Recursive: $\mathbf{t_1 = 2,\ t_n = 4t_{n-1}}$.

11*. The first three terms of a GP add to $\frac{13}{12}$ and multiply to −1. Find r and the terms.

Take the terms as $\frac{a}{r}$, a, ar. Product $= a^3 = -1 \Rightarrow a = -1$.

Sum: $-\frac{1}{r} - 1 - r = \frac{13}{12} \Rightarrow r + \frac{1}{r} = \frac{-25}{12} \Rightarrow 12r^2 + 25r + 12 = 0 \Rightarrow (4r + 3)(3r + 4) = 0$.

$r = \frac{-3}{4}$ gives $\mathbf{\frac{4}{3}, -1, \frac{3}{4}}$; $r = \frac{-4}{3}$ gives $\mathbf{\frac{3}{4}, -1, \frac{4}{3}}$ (the same numbers in reverse).

12*. If the 4th, 10th and 16th terms of a GP are x, y, z, prove that x, y, z are in GP.

$x = ar^3$, $y = ar^9$, $z = ar^{15}$. Then $\frac{y}{x} = r^6$ and $\frac{z}{y} = r^6$.

The ratios are equal (r^6), so **x, y, z are in GP** (equivalently $y^2 = xz$).

13*. Three terms of a GP add to 26 and their squares add to 364. Find them.

Terms a, ar, ar^2. $a(1 + r + r^2) = 26$ ... (1) and $a^2(1 + r^2 + r^4) = 364$ ... (2).

Use $1 + r^2 + r^4 = (1 + r + r^2)(1 - r + r^2)$. Dividing (2) by (1): $a(1 - r + r^2) = 14$ ... (3).

(1) − (3): $2ar = 12 \Rightarrow ar = 6$. (1) + (3): $2a(1 + r^2) = 40 \Rightarrow a + ar^2 = 20 \Rightarrow \frac{6}{r} + 6r = 20 \Rightarrow 3r^2 - 10r + 3 = 0 \Rightarrow r = 3$ or $\frac{1}{3}$.

The terms are **2, 6, 18** (or 18, 6, 2). Check: $2 + 6 + 18 = 26$ and $4 + 36 + 324 = 364$ ✅

14*. **$P_1 = 1$, $P_2 = 2$, $P_n = P_1 + \ldots + P_{n-1} + 1$. Find P_1 to P_8 and simpler formulas.**

$P_3 = 1 + 2 + 1 = 4$, $P_4 = 1 + 2 + 4 + 1 = 8$, and so on: **1, 2, 4, 8, 16, 32, 64, 128**.

Since $P_n = (P_1 + \ldots + P_{n-2} + 1) + P_{n-1} = P_{n-1} + P_{n-1}$, the simpler rule is $\mathbf{P_n = 2P_{n-1}}$, and the explicit rule is $\mathbf{P_n = 2^{n-1}}$.

15*. **$W_1 = 1$, $W_2 = 2$, $W_n = W_1 + \ldots + W_{n-2} + 2$. Find W_1 to W_8.**

$W_3 = 1 + 2 = 3$, $W_4 = 1 + 2 + 2 = 5$, $W_5 = 1 + 2 + 3 + 2 = 8$, $W_6 = 13$, $W_7 = 21$, $W_8 = 34$.

1, 2, 3, 5, 8, 13, 21, 34: the **Virahānka–Fibonacci sequence**! (Subtracting consecutive rules gives $W_n - W_{n-1} = W_{n-2}$.)

Sheet

Formula sheet and chapter summary

Sequence

Ordered list of numbers. t_n = term in position n.

Explicit rule

t_n in terms of n, e.g. $t_n = 2n - 1$. Is x a term? Solve $t_n = x$; n must be a natural number.

Recursive rule

Start value + rule using earlier terms, e.g. $V_n = V_{n-1} + V_{n-2}$.

AP

a, a+d, a+2d, ...

$\mathbf{t_n = a + (n - 1)d}$

$t_n = t_{n-1} + d$

GP

a, ar, ar^2, ...

$\mathbf{t_n = ar^{n-1}}$

$t_n = r \times t_{n-1}$

Sum 1 to n

$\mathbf{S_n = \frac{n(n+1)}{2}}$ = nth triangular number.

$m + \ldots + n = S_n - S_{m-1}$

Graphs

AP points: straight line.
GP points: a curve.

Two unknowns

Given two terms of an AP, write $a + (p-1)d$ and $a + (q-1)d$, subtract to get d.

Fractals

Sierpiński triangle: 3^n triangles, area $(\frac{3}{4})^n$.

Carpet: 8^n squares, area $(\frac{8}{9})^n$.

📝 Chapter summary

- A sequence is an ordered list of numbers; each number is a term.
- An explicit formula uses the position n to calculate the term's value.
- A recursive formula gives a term using the values of previous terms.
- Triangular numbers 1, 3, 6, 10, 15, ...: $t_n = \frac{n(n+1)}{2}$, which is also the sum of the first n natural numbers.
- An AP adds a fixed number d (common difference): $t_n = a + (n - 1)d$.
- A GP multiplies by a fixed number r (common ratio): $t_n = ar^{n-1}$.
- Fractals such as the Sierpiński triangle and the Sierpiński carpet lead to GPs.

Quiz

Test yourself

Tap an answer. You get one try per question.

1. What is the 20th term of $t_n = 4n - 3$?

- 80
- **77**
- 83
- 73

$4 \times 20 - 3 = 77$.

2. Which of these is an AP?

- 1, 2, 4, 8, ...
- 1, 4, 9, 16, ...
- **10, 7, 4, 1, ...**
- 1, 1, 2, 3, 5, ...

The difference is always −3.

3. Common ratio of 81, 27, 9, 3, ...?

3	−54
$\frac{1}{3}$	−3

$27 \div 81 = \frac{1}{3}$.

4. 1 + 2 + 3 + ... + 40 = ?

800	**820**
840	1600

$\frac{40 \times 41}{2} = 820$.

5. Is 100 a term of $t_n = 3n + 2$?

Yes, the 33rd	**No**
Yes, the 34th	Yes, the 98th

$3n = 98$ gives $n = 32.67$, not a natural number.

6. Next Virahānka number after 13, 21, 34?

47	68
55	42

21 + 34 = 55.

7. $t_1 = 2$, $t_n = 3t_{n-1}$. What is t_4?

- 18
- **54**
- 11
- 162

2, 6, 18, 54.

8. Plotting (n, t_n) for an AP gives...

- **A straight line**
- An upward curve
- A circle
- Random dots

Each step adds the same d, so the points line up.

9. Black triangles at Stage 4 of the Sierpiński triangle?

- 12
- 64
- **81**
- 27

$3^4 = 81$.

10. In an AP, $a = 5$ and $t_6 = 30$. What is d?

- 6
- **5**
- 25
- 4

$5 + 5d = 30 \Rightarrow d = 5$.

Notes based on NCERT Ganita Manjari (Grade 9 Mathematics, Part I), Chapter 8: Predicting What Comes Next: Exploring Sequences and Progressions. Diagrams are taken from the NCERT textbook chapter.